

PANJGUR, Pakistan

WMO: 417390

Lat: 26.955N

Lon: 64.133E

Elev: 1003

StdP: 89.85

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	2.4	-12.3	1.5	15.2	-9.5	1.9	13.8	12.0	21.7	10.1	17.8	2.6	0	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.9	41.4	21.5	40.3	21.3	39.2	21.1	25.5	36.0	24.6	35.5	23.8	34.8	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.8	19.9	31.2	21.6	18.4	30.7	20.7	17.4	31.1	84.7	36.3	80.8	35.3	77.3	34.9	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.8	7.5	-1.2	43.5	1.5	1.2	-2.3	44.4	-3.2	45.1	-4.1	45.8	-5.2	46.7		
			-3.9	26.5	1.8	0.9	-5.1	27.1	-6.2	27.7	-7.1	28.2	-8.4	28.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	11.7	13.9	19.7	26.1	30.5	32.8	33.0	31.1	28.5	24.3	17.7	12.5	
	DBStd	8.25	3.09	4.32	3.96	3.08	2.48	2.09	1.55	1.64	2.03	2.61	3.55	3.31	
	HDD10.0	51	19	15	0	0	0	0	0	0	0	0	1	16	
	HDD18.3	604	205	131	32	1	0	0	0	0	0	0	53	182	
	CDD10.0	4989	73	124	302	482	634	684	714	656	554	442	232	92	
	CDD18.3	2501	0	7	75	233	376	434	456	397	304	184	34	0	
	CDH23.3	36274	6	112	832	2912	5642	6950	7192	5927	4124	2177	383	18	
	CDH26.7	21864	0	13	289	1518	3537	4648	4761	3678	2346	996	77	1	
Wind	WSAvg	2.8	2.8	3.0	2.9	2.8	2.8	3.6	3.1	2.7	2.3	2.5	2.7	2.9	
Precipitation	PrecAvg	88	15	16	15	8	3	2	16	8	2	1	2	9	
	PrecMax	200	73	97	69	67	22	25	114	54	25	42	42	83	
	PrecMin	9	0	0	0	0	0	0	0	0	0	0	0	0	
	PrecStd	47	19	21	18	12	6	4	26	14	6	6	6	17	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.0	28.4	33.9	38.0	41.8	42.7	42.6	41.7	38.4	36.3	30.6	25.3	
		MCWB	12.9	14.4	18.0	19.6	21.9	21.2	22.1	21.4	19.7	18.7	14.9	12.9	
	2%	DB	22.2	26.4	31.8	36.2	40.1	41.5	41.1	39.6	37.1	34.6	28.9	23.3	
		MCWB	11.9	13.8	17.2	18.9	21.3	21.4	21.5	20.9	19.5	18.1	14.9	12.1	
	5%	DB	20.8	24.6	30.0	35.0	38.8	40.2	40.0	38.3	36.2	33.2	27.3	21.8	
		MCWB	11.1	13.2	16.5	18.5	21.6	21.3	21.3	20.2	19.2	17.6	14.5	11.5	
	10%	DB	19.3	22.6	28.0	33.6	37.5	39.1	38.9	37.3	35.3	31.6	25.6	20.1	
		MCWB	10.2	12.2	15.6	17.9	20.9	20.9	21.2	20.0	19.0	16.8	14.0	10.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	16.5	20.5	21.5	27.7	25.9	26.1	26.5	24.5	21.8	17.6	15.2
			MCDB	21.5	26.0	31.8	34.5	36.4	38.0	36.3	34.3	34.6	32.2	26.7	21.3
2%		WB	13.4	14.8	18.5	20.5	25.0	24.9	25.1	24.8	23.0	20.1	16.5	13.5	
		MCDB	20.6	24.2	29.3	33.5	36.5	36.6	35.8	33.9	33.9	31.8	25.8	21.2	
5%		WB	12.2	13.7	17.3	19.7	23.5	24.0	24.4	23.4	21.9	19.1	15.5	12.4	
		MCDB	19.5	22.9	27.8	32.6	36.1	36.2	35.1	32.8	32.6	30.9	25.3	20.3	
10%		WB	10.9	12.6	16.2	18.9	22.1	23.0	23.7	22.5	20.9	18.0	14.5	11.2	
		MCDB	18.0	21.5	26.3	31.4	35.1	34.9	34.3	32.3	31.8	29.7	24.5	19.0	
Mean Daily Temperature Range	5% DB	MDBR	13.8	13.5	13.8	13.9	14.0	12.9	12.7	13.2	13.7	14.5	13.7	13.9	
		MCDBR	16.0	15.6	15.8	15.4	15.0	14.1	14.1	14.2	14.6	15.7	15.2	15.8	
		MCWBR	8.9	7.9	6.8	5.7	5.8	4.8	4.6	4.7	5.4	6.2	7.1	8.7	
	5% WB	MCDBR	14.4	14.1	15.0	14.5	14.3	12.9	12.6	12.9	13.8	15.2	14.1	14.7	
		MCWBR	8.9	8.3	7.4	6.1	7.4	5.6	5.0	5.3	6.0	7.0	7.4	9.0	
Clear-Sky Solar Irradiance	taub	0.336	0.384	0.433	0.499	0.534	0.562	0.632	0.620	0.560	0.460	0.398	0.350	0.327	
	taud	2.332	2.178	2.011	1.867	1.796	1.750	1.654	1.654	1.774	1.946	2.098	2.279	2.353	
	Ebn at Noon	907	883	858	810	780	754	702	751	821	855	882	901		
	Edh at Noon	109	138	173	205	221	230	252	222	182	147	114	103		
All-Sky Solar Radiation	RadAvg	4.19	4.97	6.08	6.92	7.42	7.56	7.01	6.85	6.51	5.69	4.68	4.04		
	RadStd	0.25	0.30	0.25	0.26	0.17	0.28	0.33	0.26	0.17	0.13	0.13	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page